D(QCOH) EQUALS D_QCOH FOR NOETHERIAN ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author environment labelled proposition-Noetherian; the proof environment immediately following it is its own proof

Editorial note. The label proposition-Noetherian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Allowed standard prerequisites (editorial).

- the pushforward lemma for Noetherian spaces (lemma-Noetherian-pushforward of the same chapter)
- the argument lemma of the same chapter (lemma-argument-proves) establishing the quasi-inverse

Source reference. Original equation 05NN: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 509–509. Prerequisite pointer invoked at 3037; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09T3: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2966–2966. Prerequisite pointer invoked at 3047; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09T6: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2490–2490. Prerequisite pointer invoked at 3048; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Then the functor (05NN)*

$$D(QCoh(\mathcal{O}_X)) \longrightarrow D_{QCoh}(\mathcal{O}_X)$$

is an equivalence with quasi-inverse given by RQ_X.

Proof. Follows immediately from Lemmas 09T3 and 09T6. □

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